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20 March 1986

SUPERSEDING
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MILITARY SPECIFICATION

GREASE, SILICONE FOR USE WITH AMMUNITION

This specification is approved for use within US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers one type of silicone grease for use in ammunition (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense index of Specifications and Standards (DoDISS) specified in the solication from a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-C-502	-Copper Rods and Shapes; and Flat Products with Finished Edges (Flat Wire, Strips and Bars)

STANDARDS

Federal

FED-STD-791	-Lubricants, Liquid Fuels, and Related Products; Methods of Testing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament, Research and Development Center, ATTN: SMCAR-ESC-AS, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1375

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Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-290	- Packaging, Packing and Marking of Petroleum and Related Products
MIL-STD-650	- Explosive, Sampling, Inspection Testing

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the document which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM

ASTM D 942	Greases, Lubricating, Oxygen Stability Of, BY The Oxygen Bomb Method
ASTM D 972	Greases and Oils, Lubricating, Evaporation Loss Of
ASTM D 1264	Greases, Lubricating, Water Washout Characteristics Of
ASTM D 4057	Manual Sampling of Petroleum Products
ASTM D 4177	Automatic Sampling of Petroleum and Petroleum Products

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

2.3 Order or precedence. In the event of a conflict between the test of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Composition. The grease shall be a smooth homogeneous mixture of methyl phenyl silicone fluid and a lithium soap free from any abrasive or

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other undesirable fillers as impurities. Minor proportions of additives to improve exudation stability, rust inhibition, mechanical stability, texture and wear resistance may be included.

3.2 Appearance. The grease shall be completely homogeneous and shall be of a smooth consistency, free from lumps or granulation, when tested as specified in 4.6.1.

3.3 Odor. The grease shall be free from any rancidity or perfume odor when tested as specified in 4.6.2.

3.4 Corrosion resistance. The grease, when tested with copper, shall show no decomposition, change in color, or consistency. The copper strip shall show no evidence of green or black discoloration, etching or pitting. A brown stain on the copper strip shall not be considered as evidence of corrosion when tested as specified in 4.6.3.

3.5 Viscosity. The viscosity shall comply with Table I when tested as specified in 4.6.4. Indicated temperature shall be as indicated ± 1°C (± 2°F).

Degrees	Viscosity (Centipoise)
25°C (77°F)	80,000 to 110,000
74°C (165°F)	60,000 to 100,000
-54°C (-65°F)	250,000 Maximum (Max.)

TABLE I. VISCOSITY

3.6 Specific gravity. The specific gravity at 25± 1°C (77±2°F) for 24 hours shall be 0.99 ± 0.02 when tested as specified in 4.6.5.

3.7 Evaporation. The evaporation at 93 ±2°C (200 ± 5°F) for 24 hours shall be no more than 2 percent, max., when tested as specified in 4.6.6.

3.8 Bleeding. The bleeding at 93 ± 2°C (200 ± 5°F) for 24 hours shall be no more than 25 percent, max., when tested as specified in 4.6.7.

3.9 Oxidation stability. The oxidation stability at 93 ± 2°C (200 ± 5°F) for 100 hours shall be no more than 5 psi max. when tested as specified in 4.6.8.

3.10 Water resistance. The water resistance at 38 ± 1°C (100 ± 2°F) shall be no more than 10 percent max., when tested as specified in 4.6.9.

3.11 Dirt count. The dirt count of the grease shall not exceed the amount specified in Table II when tested as specified in 4.6.10.

Microns or above	Per cubic centimeter Max.
25	7500
75	1600
125	None

TABLE II. DIRT COUNT

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Submission of product. At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawings, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual document.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Lot.

4.2.1 Production lot. An indefinite quantity of a homogeneous mixture of product offered for acceptance in a single, isolated container, or manufactured in a single plant run, through the same processing equipment, with no change in the ingredient materials in a 24-hour time period.

4.2.2 Packaged lot. An indefinite number of unit containers of identical size and type, offered for acceptance at one time, and filled with a homogeneous mixture of product manufactured in a single plant run, through the same processing equipment, in a 24-hour time period.

4.3 Sampling.

4.3.1 Sampling for examination of filed containers. Take a random sample of filled containers from each lot in accordance with MIL-STD-105 at inspection level II and acceptable quality level (AQL) - 2.5 percent defective.

4.3.2 Sampling for tests. Take samples for tests in accordance with ASTMs D4057 or D4177, as applicable.

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4.4 Inspection.

4.4.1 Inspection of material. Perform inspection of material in accordance with method 9601 of Federal Test Method Standard No. 791.

4.4.2 Examination of filled containers. Examine samples taken in accordance with 4.3.1 for compliance with MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing and marking requirements. Reject any container having one or more defects or under the required fill. If the number of defective or underfilled containers exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, reject the lot represented by the sample. Container material shall be of a density sufficient to prevent migration of the product or any constituents through the container. Container will be pretreated to insure that printed identification will properly adhere and will not be removed by the product.

4.5 Quality conformance inspection. Quality conformance inspection consist of the tests indicated in Table III performed on samples taken in accordance with 4.3.2. If a product sample fails to meet any of the tests specified in Table III, the lot represented by the sample shall be rejected.

4.6 Methods of examinations and tests. Perform tests in accordance with Table III.

TABLE III - QUALITY CONFORMANCE TESTS

INSPECTION	PARAGRAPH	
	REQUIREMENT	TEST METHOD
Appearance	3.2	4.6.1
Odor	3.3	4.6.2
Corrosion Resistance	3.4	4.6.3
Viscosity	3.5 (Table I)	4.6.4
Specific Gravity	3.6	4.6.5
Evaporation	3.7	4.6.6
Bleeding	3.8	4.6.7
Oxidation Stability	3.9	4.6.8
Water Resistance	3.10	4.6.9
Dirt Count	3.11 (Table II)	4.6.10

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4.6.1 Appearance, Code No. (See 6.3) 01001. A suitable portion of the grease shall be worked with a spatula on a clean glass. During working, the grease shall show no nonuniformity and shall spread with a straight edge to a smooth, glossy, level surface.

4.6.2 Odor, Code No. 02001. A nasal analysis shall be made of the grease.

4.6.3 Corrosion resistance, Code No. 03001. Prepare a strip of clean copper conforming to QQ-C-502, 1.75 x 0.25 x 0.02 inch, hard temper, bent midway between the ends at an angle of 45°. Dip the copper strip for 15 to 20 seconds into an aqueous solution containing 25% by weight sulfuric acid and 25% by weight nitric acid. Wash immediately in distilled water and give a bright dip for 15 seconds in an aqueous solution of approximately 2% sulfuric acid by weight and 5% potassium dichromate by weight. Wash well in distilled water and dry for 30 minutes in an oven maintained at approximately 100 ± 2°C (212 ±5°F). (The metal must not be touched with the fingers at any time during or after cleaning). Remove the copper strip from the oven and cool to room temperature. Place four grams of sample grease in a glass dish, insert the copper strip horizontally into the grease so that one-half of the copper strip is immersed and place the test dish into a Norma-Hoffman at 99 ± 2°C (210 ± 5°F), and 110 ± 0.25 pounds per square inch (p.s.i.) initial oxygen pressure. At the end of 20 hours of test, remove the glass dish. Visually examine the copper strip and the sample of grease for conformance with 3.4.

4.6.4 Viscosity, Code No. 04001. The viscosity shall be determined with a Brookfield RVF Viscosimeter with a No. 7 spindle at 4 rmp. The spindle shall be heated or cooled in the grease, depending upon the temperature at which the viscosity is being determined. The test shall be performed in a small cold box at -54 ± 1°C (-65 ± 2°F). The sample shall not be disturbed for a least 24 hours before the test. The readings shall be taken 1 to 2 minutes after starting the motor.

4.6.5 Specific gravity, Code No. 05001. The specific gravity shall be determined in accordance with Method 203.2 of MIL-STD-650.

4.6.6 Evaporation, Code No. 06001. The evaporation shall be determined in accordance with ASTM D972 except that the time shall be 24 hours at 93 ± 2°C (200 ± 5°F).

4.6.7 Bleeding, Code No. 07001. The bleeding shall be determined in accordance with Method 321.2 of FED-STD-791, except that the test shall be conducted at 93 ± 2°C (200 ± 5°F) for 24 hours.

4.6.8 Oxidation stability, Code No. 08001. The oxidation stability shall be determined in accordance with ASTM D942, except that the test shall be conducted at 93 ± 2°C (200 ± 5°F) for 100 hours.

4.6.9 Resistance, Code No. 09001. The water resistance shall be determined in accordance with ASTM D1264 at 38 ± 1°C (100 ± 2°F).

4.6.10 Dirt count, Major defect, Code No. 11001. The dirt count shall be determined in accordance with Method 3005 of FED-STD-791.

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5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified in the contract or purchase order (see 6.2), preservation, packing and marking shall be in accordance with MIL-STD-290. Additionally, all unit and intermediate packs of toxic and hazardous chemicals and materials shall be labeled in accordance with the applicable laws, statutes, regulations or ordnances, including Federal, State, and Municipal requirements.

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use as a waterproofing agent in ordnance items such as the M605 Mine, Fuze.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Quantity of grease required.
c. Type and size container.
d. Level of preservation and packing (MIL-STD-290).

6.3 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

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